

JONATHAN TROWBRIDGE

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CAREER SUMMARY

Security Researcher & Principal Software Engineer with 10+ years of experience building secure, scalable web, mobile, and backend systems. **Certified in OSCP+, GPEN, GCIH, GSEC, and GFACT** with a deep technical foundation in offensive security and original vulnerability research resulting in 9 disclosed vulnerabilities across Debian packages and Linux system utilities, including multiple root-level privilege escalations. I've also reproduced functional PoCs for several known, critical-rated CVEs to validate real-world exploitability.

PROFESSIONAL EXPERIENCE

Independent Security Researcher | Richmond, Virginia Oct. 2025 - Present

- **Local Privilege Escalation:** discovered and disclosed a root-level privilege escalation in a widely-deployed Linux database server package via insecure symlink handling (CWE-59). Under embargo pending CVE assignment.
- **Multi-Stage Privilege Escalation Chain:** discovered and disclosed a root-level privilege escalation in a Debian build-infrastructure utility, chaining insecure permissions, path traversal, and configuration injection (CWE-732/CWE-22). Under embargo pending CVE assignment.
- **Linux Privilege Escalation via 9mount:** Discovered and reported a root exploit chain in the SUID-root 9mount utility; fixed in 9mount 1.5
- **sohps (Shared Object Hijack Path Scanner):** Developed a security auditing tool to identify shared object hijack vulnerabilities in Linux ELF binaries and AppImage containers. Used it to scan ~70,000 Debian packages, resulting in 6 disclosed vulnerabilities: Debian Bugs [#1138498 \(python3-torch\)](#), [#1138589 \(libarrow500\)](#), [#1138592 \(gpu-basis-universal\)](#), [#1138596 \(python3-ifcopenshell\)](#), [#1138597 \(python3-ledger\)](#), and [#1138072 \(clang-tools-21\)](#)
- Developed functional PoCs reproducing 4 critical-rated CVEs: [CVE-2026-21858 \(ni8mare\)](#), [CVE-2025-55182 \(React2Shell\)](#), [CVE-2026-33017 \(Langflow\)](#), [CVE-2026-1357 \(WPVivid\)](#)

Lead Principal Software Engineer | Coinme | Seattle, WA (Remote) Sept. 2021 - Oct. 2024

- Directed the native platform migration to Flutter, architecting authentication and payment flows to ensure PCI data security and transaction integrity.
- Engineered a secure mobile architecture using Riverpod, implementing state-management patterns to prevent PII leakage across modular, decoupled feature sets.
- Established a secure SDLC and mentored team members to accelerate the delivery of high-impact features while ensuring architectural integrity and scalability.

Senior Full Stack Software Engineer | CoStar Group | Richmond, VA (Hybrid) Jan. 2019 - Aug. 2021

- Led development of an Android listing module and supporting ASP.NET Core API, ensuring secure data handling and endpoint protection for high-traffic commercial services.
- Engineered scalable web applications with React and Node.js and built an enterprise AWS push notification system for secure, real-time cross-platform communication.
- Partnered cross-functionally to implement secure design principles and code review standards, ensuring resilient and consistent feature releases across mobile and web.

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Full Stack Software Engineer | Mapleton Hill Media | Boulder, Colorado

Aug. 2016 – Feb. 2018

- Developed cross-platform mobile applications for logistics and agriculture, ensuring secure client-side data handling and architectural consistency.
- Modernized legacy ASP.NET architectures by migrating monolithic APIs to modular services, improving service isolation and system auditability.
- Engineered real-time synchronization for distributed inventory systems, implementing secure data flows and validation logic across mobile devices and APIs.

Full Stack Application Developer | Worldview Solutions Inc. | Richmond, VA

Feb. 2013 – Feb. 2016

- Led implementation of claims-based authentication and IAM for a U.S. Air Force recruiting portal, ensuring strict access control and federal compliance.
- Developed regulatory and monitoring modules for Virginia state agencies, focusing on data integrity and workflow efficiency within ASP.NET environments.
- Modernized legacy web architectures to improve performance and maintainability, ensuring architectural consistency across client projects.

Software Developer | Gallium Technology | Richmond, VA

2012 - 2013

Quality Assurance Internship | Prematics Inc. | McLean, VA

2008

CYBERSECURITY CERTIFICATIONS & ACHIEVEMENTS

RVAssec 2026 CTF | 3rd place, individual category

2026

OSCP+ | Offensive Security Certified Professional+, OffSec

2025

GPEN | GIAC Penetration Tester, SANS Technology Institute

2025

GCIH | GIAC Certified Incident Handler, SANS Technology Institute

2025

GSEC | GIAC Security Essentials, SANS Technology Institute

2025

GFACT | GIAC Foundational Cybersecurity Technologies, SANS Technology Institute

2024

TECHNICAL SKILLS

Security Research & Engineering: Penetration Testing, Vulnerability Research, Exploit Development, Web & Mobile Application Security, Network Security, API Security, Threat Modeling, Security-Focused Code Review

Languages: Python, Bash, PowerShell, Kotlin, Java, C#, TypeScript, Dart, Go

Frameworks: Flutter, KMP, Xamarin, Ktor, ASP.NET Core, Node.js, React, Angular

Infrastructure & Environments: Linux, Windows Server, Docker, Git, CI/CD Pipeline Security

Databases: SQL Server, MariaDB, Redis, PostgreSQL, SQLite, MongoDB

EDUCATION

Undergraduate Certificate in Applied Cybersecurity | SANS Technology Institute

2025

Bachelor of Science in Computer Science | Virginia Commonwealth University

2013

Expanded resume: masterwok.github.io | Additional project details and private repositories available upon request.